Optimization

maximise
minimise

what is it to say optimize?

optimize - What are we optimizing?
How do we optimize

Goal
↓
objective fn
↓
cost fn

To find out the value of the parameters s.t we optimize the objective fn

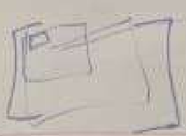
a) optimise a fn without any restrictions on the parameters → unconstrained optimizn.

b) optimize a fn with restrictions on the parameters → constrained optimizn

* Maxima / Minima → Extrema

suppose $y = f(x)$ then $\frac{dy}{dx} = 0$ gives you the point at which the slope is zero

Sign of $\left(\frac{d^2y}{dx^2}\right)$ at the pt $\frac{dy}{dx} = 0$ tells us the nature of extrema



→ minor?

a) If $D > 0$ & $f_{xx}(x_0, y_0) > 0$, then f has a relative min at (x_0, y_0)

b) If $D > 0$ & $f_{xx}(x_0, y_0) < 0$, then f has a relative max at (x_0, y_0)

c) If $D < 0$, then f has a saddle pt at (x_0, y_0)

d) If $D = 0$, we cannot conclude!

Ex: $f(x,y) = 3x^2 - 2xy + y^2 - 8y$. Find all relative extrema & saddle point of f.

soln: $f_x(x,y) = \frac{\partial f}{\partial x}(x,y) = 6x - 2y$

$f_y(x,y) = \frac{\partial f}{\partial y}(x,y) = 2y - 2x - 8$

$6x = 2y$

$3x = y$

$(2,6)$

$6x - 2x - 8 = 0$

$4x = 8$

$x = 2$

$y = 6$

$$H = \begin{bmatrix} 6 & -2 \\ -2 & 2 \end{bmatrix} \Rightarrow Det(H) = 12 - 4 = 8 > 0$$

$\therefore$ f has a rel. min at $(2,6)$

$$1 = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx$$

$$\Rightarrow \int_{-\infty}^{\infty} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx = \sqrt{2\pi}\,\sigma$$

$$\mu = 0 ; \sigma = 1 \Rightarrow \sqrt{2\pi} = \int_{-\infty}^{\infty} e^{-x^2/2} dx$$

$$\int_{-\infty}^{\infty} x e^{-x^2/2} dx = E[X] = \mu$$

$$= 0$$

$$\Rightarrow \int_{-\infty}^{\infty} x e^{-x^2/2} dx = 0$$

Optimiz^n:

$f : R^n \to R \qquad y = f(\vec{x}) = f(x_1, x_2 \ldots x_n)$

$$\frac{\partial f}{\partial \vec{x}} = \left[\frac{\partial f}{\partial x_1} \quad \frac{\partial f}{\partial x_2} \quad \cdots \quad \frac{\partial f}{\partial x_n} \right]$$

must be continuous

Equate partial derivatives to 0 & get the values of $x_1 \ldots x_n$

$(x_1^0, x_2^0, \ldots x_n^0)$

Also compute Hessian $= \nabla^2 f = \frac{\partial}{\partial \vec{x}} \left(\frac{\partial f}{\partial \vec{x}} \right)$

Det of $H > 0$; $\frac{\partial^2 f}{\partial x_1^2} > 0 \rightarrow$ Min at the point(s)

$H > 0$; $\frac{\partial^2 f}{\partial x_1^2} < 0 \rightarrow$ Max at the point(s)

$D < 0$ saddle point

$D = 0$ No conclusion

$0 \le y \le 5$

$\frac{dv}{dy} = -3$

$(0,5)$ & $(3,0)$

$5x + 3y = 15$

$y = \frac{-5x}{3} + 5$

$0 \le x \le 3$

$f(x,y) = f(x, \frac{-5x}{3} + 5)$

$= 3xy - 6x - 3y + 7$

$= 3x\left(\frac{-5x}{3} + 5\right) - 6x - 3\left(\frac{-5x}{3} + 5\right) + 7$

$= -5x^2 + 15x - 6x + 5x - 15 + 7$

$= -5x^2 + 14x - 8$

$f'(x) = -10x + 14 = 0 \Rightarrow x = +7/5$

$y = \frac{-5x}{3} + 5 = \frac{-5(7/5)}{3} + 5$

$-18 \quad = 8/3$

(x,y)	$(0,0)$	$(3,0)$	$(0,5)$	$(1,2)$ $(7/5, 8/3)$
$f(x,y)$	7	-11	-8	, ,

$\cancel{(1,2)}$ & $\cancel{(-1,-2)}$ $(\pm 1, \pm 2)$

$$H = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad D = -1$$

$f(1,2) = 2$

$f(-1,2) = f(1,-2) = -2$

$f(-1,-2) = 2$

Q) $f = 2x_1 + 3x_2 + x_3 + x_4 - x_5$,

$x_1^2 + x_2^2 + x_3^2 + x_4^2 + x_5^2 - 1 = 0$

$\Rightarrow \nabla f = [2 \quad 3 \quad 1 \quad 1 \quad -1]$

$\nabla g = [2x_1 \quad 2x_2 \quad 2x_3 \quad 2x_4 \quad 2x_5]$

$\Rightarrow \frac{1}{x_1} = \lambda, \quad \frac{3}{2x_2} = \lambda, \quad \frac{1}{2x_3} = \lambda, \quad \frac{1}{2x_4} = \lambda, \quad \frac{-1}{2x_5} = \lambda$

$\Rightarrow \frac{1}{\lambda^2} + \frac{9}{4\lambda^2} + \frac{1}{4\lambda^2} + \frac{1}{4\lambda^2} + \frac{1}{4\lambda^2} - 1 = 0$

$\Rightarrow \frac{4 + 9 + 3}{4\lambda^2} = 1 \Rightarrow \lambda^2 = 16$

$\lambda = \pm 2$

$\Rightarrow x_1 = \pm \frac{1}{2}, \quad x_2 = \pm \frac{3}{8},$

$x_3 = x_4 = \pm \frac{1}{8}, \quad x_5 = \mp \frac{1}{8}$

$\frac{1}{z} = \lambda - \frac{2}{y}$ $\quad \frac{1}{z} = \lambda - \frac{2}{x}$

$\lambda - \frac{2}{y} = \lambda - \frac{2}{x} \Rightarrow y = x$

$\Rightarrow \frac{1}{z} = \frac{\lambda y - 2}{y} \Rightarrow$

$z = \frac{y}{\lambda y - 2}$

$\Rightarrow z = \frac{x}{2} \quad \Rightarrow xyz = 32$

$\Rightarrow \frac{x^3}{2} = 32$

$x^3 = 64$

$x = 4, y = 4, z = 2$

$f(x,y) = xy;\ g(x,y) = 4x^2 + y^2 - 8 = 0$

Find max & min, subject to g, if they exist

$\nabla f = [y \quad x]$, $\nabla g = [8x \quad 2y]$

$y = 8\lambda x, \quad x = 2\lambda y$

$y = 8\lambda(2\lambda y) \Rightarrow \frac{1}{16} = \lambda^2$

$\lambda = \pm\frac{1}{4}$

$\Rightarrow y = 2x, \quad x = \frac{y}{2}$

$\Rightarrow 4\left(\frac{y^2}{4}\right) + y^2 - 8 = 0$ $\quad y = \pm 2$

$y^2 = 4$ $\quad x = \pm 1$

$\lambda \rightarrow$ lagrange multiplier

* use lagrange multipliers to det the dim of a rectangular box open at top, $V = 32$, requiring least amount of material

soln : $f = xy + 2yz + 2xz$

$xyz = 32$

$\nabla f = \begin{bmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} & \frac{\partial f}{\partial z} \end{bmatrix}$

$= \begin{bmatrix} y + 2z & x + 2z & 2x + 2y \end{bmatrix}$

$\nabla g = \begin{bmatrix} yz & xz & xy \end{bmatrix}$

$\boxed{\text{at the constrained min, } \nabla f = \lambda \nabla g}$

$\Rightarrow \begin{bmatrix} y+2z & x+2z & 2x+2y \end{bmatrix} = \lambda \begin{bmatrix} yz & xz & xy \end{bmatrix}$

$\Rightarrow y + 2z = \lambda yz$

$x + 2z = \lambda xz$

$2x + 2y = \lambda xy$

also since box, $x > 0, y > 0, z > 0$

$\therefore \frac{1}{z} + \frac{2}{y} = \lambda \quad \frac{1}{z} + \frac{2}{x} = \lambda \quad \frac{2}{y} + \frac{2}{x} = \lambda$

Lagrange multipliers:

$z = f(x, y)$ constraint: $g(x, y) = 0$

Let $f(x,y)$ be the objective fn to be maximised subject to the constraint $g(x, y) = 0$

~~Let $f(x,y)$ be the objective fn to be~~

By saying maximize w.r.t to a constraint, we want to find out (x_0, y_0) on the constraint curve s.t $f(x,y)$ is max

$f(x,y) \rightarrow$ objective fn $\quad x, y \quad f(x,y) = c_1$

$x, y \quad c_2$

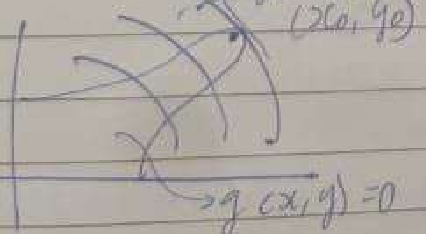
c_3

Level curves

$\Rightarrow$ To locate such a point, on the constraint curve that maximise/min f, we look the points of intersec^n of $g(x, y) = 0$ with a level curve

At a point (x_0, y_0) both $f(x,y)$ & $g(x,y) = 0$

share a common tangent



$\nabla f(x_0, y_0) \Rightarrow \perp$ to the level curve

$\nabla g(x_0, y_0) \Rightarrow \perp$ to the constraint curve

$\vec{\nabla} f(x_0, y_0) \parallel \vec{\nabla} g(x_0, y_0)$

$\Rightarrow \vec{\nabla} f(x_0, y_0) = \lambda \vec{\nabla} g(x_0, y_0)$

$$\frac{\partial s}{\partial x} = 2x - 2(27-x-y)$$

$$x = 27-x-y$$

$$x = \frac{27-y}{2}$$

$$\frac{\partial s}{\partial y} = 2y - 2(27-x-y)$$

$$y = \frac{27-x}{2}$$

$$y = \frac{27}{2} - \frac{(27-y)}{4}$$

$$y = \frac{54-27+y}{4}$$

$$4y = 27+y \qquad 3y = 27$$

$$y = 9$$

$$x = 9$$

$H = \begin{bmatrix} 4 & 2 \\ 2 & 4 \end{bmatrix}$, min at $(9,9,9)$

Min/Max f, constraint c

Is it possible to convert a constrained optimization problem to an unconstrained one?

ex: Volume open box problem, $V=32$

$s = xy + 2yz + 2zx \qquad f(x,y,z) = s$

$xyz = 32 \qquad g(x,y,z) = c$

$\Rightarrow xyz - 32 = 0 \rightarrow g(x,y,z) = 0$

$$z = \frac{v}{xy} = \frac{v}{(\sqrt[3]{2v})(\sqrt[3]{2v})}$$

$$= \frac{v}{(2v)^{2/3}} = \frac{1}{2}(\sqrt[3]{2v})$$

Find $\frac{\partial^2}{\partial x^2}$, $\frac{\partial^2}{\partial y^2}$, $\frac{\partial^2}{\partial x \partial y}$

$$\frac{\partial^2 s}{\partial x^2} = \frac{4v}{x^3}\Big|_{at\ (\sqrt[3]{2v}, \sqrt[3]{2v})} = 2$$

$$\frac{\partial^2 s}{\partial y^2} = \frac{4v}{y^3}\Big|_{(\sqrt[3]{2v}, \sqrt[3]{2v})} = 2$$

$$\frac{\partial^2 s}{\partial y \partial x} = 1, \quad \frac{\partial^2 s}{\partial x \partial y} = 1$$

$$H = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \qquad D = 3 > 0$$

relative min

* Find 3 +ve no whose sum is 27 & their sum of sqs. is as small as possible.

$$x + y + z = 27 \Rightarrow S = x^2 + y^2 + z^2$$

$$z = 27 - x - y \qquad S = x^2 + y^2 + (27 - x - y)^2$$

$$S = 2x^2 + 2y^2 + 729 - 54x - 54y + 2xy$$

$$\frac{\partial s}{\partial x} =$$

ex: Det the dimensions of a rect. box, open at the top, with vol V & req. least amount of material for construction.

soln: x, y, h

$lbh = V$, min $S = lb + 2lh + 2bh$

$x > 0, y > 0, z > 0$

$v = xyz \Rightarrow z = \frac{v}{xy}$

$S = xy + \frac{2Vy}{xy} + 2\left(\frac{V}{xy}\right)x = xy + \frac{2V}{y} + \frac{2V}{x}$

$\frac{\partial S}{\partial x} = y - \frac{2V}{x^2}$, $\frac{\partial S}{\partial y} = x - \frac{2V}{y^2}$

$y = \frac{2V}{x^2}$ $\qquad x = \frac{2V}{y^2}$

$x = \frac{2V x^2}{\cancel{4V^2}\, 2V} \Rightarrow \frac{x^2}{2V} - x = 0$

$\Rightarrow x^2 - 2Vx = 0$

$\Rightarrow x(x - 2V) = 0$

$x = 2V$

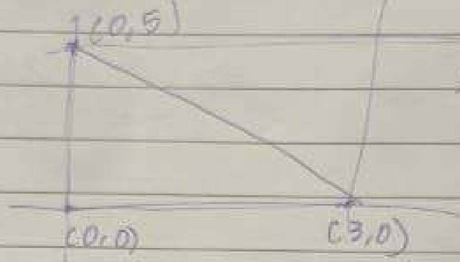
critical pt

$(\sqrt[3]{2V}, \sqrt[3]{2V})$

Learn eq of line again

* Find the abs max & min of
$f(x,y) = 3xy - 6x - 3y + 7$

on the closed region R defined by vertices (0,0), (3,0) & (0,5)



$\frac{x}{3} + \frac{y}{5} = 1$

$5x + 3y = 15$

$5 + 6 = 15$

$\frac{\partial f}{\partial x} = 3y - 6$, $\frac{\partial f}{\partial y} = 3x - 3$

$y = 2$ $\qquad$ $x = 1$

(1, 2) falls inside R

Learn ~~this method~~

R has 3 line segments

(0, 0) & (3, 0) $\quad$ (0,0) & (0,5) $\quad$ (0,5) & (3,0)

$u(x) = f(x, 0)$

$= -6x + 7, \quad 0 \le x \le 3$

$\frac{du}{dx} = -6$ Non zero for all x

$\Rightarrow \therefore$ no critical points

(0, 0) & (0, 5)

$v(y) = f(0, y)$

$= -3y + 7$

Ex:2 Locate all relative extrema & saddle points of $f(x,y) = 4xy - x^4 - y^4$

$f_x = 4y - 4x^3 \qquad f_y = 4x - 4y^3$

$\Rightarrow y = x^3 \quad , \quad x = y^3$

$(x,y) \rightarrow (0,0), (1,1), (-1,-1)$

$x^8 - 1 = 0$

use Euler's for all roots

$\Rightarrow H = \begin{bmatrix} -12x^2 & 4 \\ 4 & -12y^2 \end{bmatrix}$

$(0,0)$ $(1,1)$ & $(-1,-1)$

-16 128

saddle point

$(1,1)$ -12 $(-1,-1)$

relative maxima

∴ For now ignore imaginary roots as this fn maps $\mathbb{R}^2 \rightarrow \mathbb{R}$

$$\int_{-\infty}^{\infty} x e^{-x^2/2} \, dx$$

$$\int_{-\infty}^{\infty} e^{-x^2/2} \, dx$$

Ex: $f(x,y) = x^2 - y^2$

$$\frac{\partial f}{\partial x} = 2x \qquad \frac{\partial f}{\partial y} = -2y$$

at $(x_0, y_0) = (0,0)$ $\quad \frac{\partial f}{\partial x} = 0 \; ; \; \frac{\partial f}{\partial y} = 0$

f does not have max or min at $(0,0)$. The point $(0,0)$ here is called a SADDLE POINT

* The second partials test:
Let $f : \mathbb{R}^2 \mapsto \mathbb{R}$ with continuous second-order partial derivatives in some disk centered at the critical point (x_0, y_0) and let

Hessian matrix

$$D = f_{xx}(x_0, y_0)\, f_{yy}(x_0, y_0) - f_{xy}^2(x_0, y_0)$$

$$H = \begin{bmatrix} f_{xx}(x_0, y_0) & f_{xy}(x_0, y_0) \\ f_{x,y}(x_0, y_0) & f_{yy}(x_0, y_0) \end{bmatrix}$$

$$D = \det(H)\Big|_{(x_0, y_0)}$$

$$\nabla f = \begin{bmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} \end{bmatrix}$$

$$\nabla^2 f = H = \nabla(\nabla f) = \begin{bmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial x \partial y} & \frac{\partial^2 f}{\partial y^2} \end{bmatrix}$$

If $\frac{d^2y}{dx^2} < 0 \rightarrow$ The extrema is max

$\frac{d^2y}{dx^2} > 0 \rightarrow$ The extrema is min

$\frac{d^2y}{dx^2} = 0$ saddle point $\rightarrow$ neither a maxima nor a minima

$y = mx + c$

$\frac{dy}{dx} = \underbrace{m}_{\text{Not zero}}$, $\frac{d^2y}{dx^2} = 0$

No minima (for a line)

Suppose f is a fn of (x, y)
If f has a relative extremum at a point (x_0, y_0) and if the first order partial derivatives of ~~..~~ f exists at (x_0, y_0) then

$$f_x(x_0, y_0) = \left.\frac{\partial f}{\partial x}\right|_{(x_0, y_0)} = 0$$

$$f_y(x_0, y_0) = \left.\frac{df}{\partial y}\right|_{(x_0, y_0)} = 0$$

At point (x_0, y_0) in the domain of the fn $f(x, y)$ is called a CRITICAL POINT of f if $\frac{\partial f}{\partial x}(x_0, y_0) = 0$ &

$\frac{\partial f}{\partial y}(x_0, y_0) = 0 \Rightarrow f_y(x_0, y_0) = 0$

$f_x(x_0, y_0) = 0$